This document contains the guidelines and policies for the Capstone Project 1& 2. These were developed with the goal of supporting BSIS students from Cebu Technological University - Main Campus's College of Computer, Information, and Communications Technology (CCICT) in producing high-quality and industry-relevant capstone projects.

Before taking this course, the student must have taken and passed the following major subjects indicated below.

For Capstone Project 1:

- Introduction to Computing
- Computer Programming 1 (Lecture and Laboratory)
- Computer Programming 2 (Lecture and Laboratory)
- Fundamentals of Information Systems
- Organization and Management Concepts
- Data Structures and Algorithms (Lecture and Laboratory)
- Professional Issues and Information Systems
- Financial Management
- Computer Architecture and Organization
- Information Management (Lecture and Laboratory)
- IT Infrastructure and Network Technologies
- Business Process Management
- Web I: Client Development
- Systems Analysis and Design (Lecture and Laboratory)
- Enterprise Architecture
- Evaluation of Business Performance
- Quantitative Analysis
- Web II: Web Application Development
- Database Administration DBMS

For Capstone Project 2:

- Capstone Project 1
- IS Project Management 1
- Data Mining
- Customer Relationship Management
- Application Development and Emerging Technologies
- Foundations of Software Engineering
- System Administration and Maintenance

Together with Capstone Project 2:

- IS Strategy, Management and Acquisition
- Enterprise Systems
- IT Audit and Controls

University Vision : CTU as a premier, inclusive, globally-recognized research and innovation, smart, community-responsive, and sustainable technological university.

University Mission : The University is primarily providing leading-edge degree programs, innovative professional, entrepreneurial, and technical instruction as well as research, extension and resource generation programs that address both the needs of the region and the nation in the context of the global knowledge economy, Fifth Industrial Revolution, and sustainability.

University Goals : By the end of 2024, CTU will be globally recognized for its quality. Specifically, it will

- Obtain a Philippine university system status with a CHED SUC LeveI V recognition or equivalent;
- Be ranked among Top 10 Universities in the Philippines;
- Achieve world-class performance and excellence using leading quality management systems for quality assurance and control;
- Be an internationally recognized and ranked university (including Quacquarelli Symonds, Times Higher Education, Shanghai University Ranking, World University Rankings);
- Develop innovative academic initiatives, including medical and allied fields engineering and technology, and transnational higher education programs for enhance career pathways, qualifications, recognitions, accreditations, specialization and progression programs, and global mobility; and
- Lead innovations in the Fifth Industrial Revolution.

University Outcomes :

1. Transformed visionary leadership in innovation to accelerate regional and national development in the context of the modern industrial revolution;
2. Increased access by economically challenged but deserving student to relevant, quality, and world-class tertiary education with learning integration of practical experiences to achieve inclusive and sustainable growth;
3. Developed research-intensive and multi-disciplinary university culture that strengthens higher education research, leading to economic productivity and innovation;
4. Instituted innovative systems resulting in effective, efficient, sustainable and equitable resource general, mobilization, and management, including optimal utilization of resources provided by the government and expansion of the other revenue and funding streams for financial sustainability;
5. Heightened sustainable community and stakeholder engagements, with harmonized institutional, financial, administrative, and policy frameworks that attain the highest academic, research, extension and resource generation excellence standards;
6. Forged strategic local and international linkages, partnerships, and initiatives leading to heightened regional and global role in the modern industrial revolution and human and intellectual capital for sustained economic development.

Quality Policy	:	CTU in compliance with statutory and regulatory requirements is committed to provide quality instruction, research, extension and production toward customer satisfaction.
Core Values	:	Commitment, Transparency, Unity, Patriotism, Integrity, Excellence, Spirituality (CTU PIES)
Graduate Attributes	:	Highly skilled individual, ethically-imbued professional, service-oriented, and effective communicator
Program Educational Objectives	:	PE01 A graduate of Information Systems will showcase advanced and innovate systems integration proficiency, designing information systems that leverage cutting-edge technologies for seamless functionality and efficiency in diverse organizational contexts. PE02 A graduate of Information Systems will excel in inclusive information management, making informed decisions based on data analytics to ensure that information systems meet the needs of diverse stakeholders, fostering inclusivity and accessibility. PE03 A graduate of Information Systems will apply their skills to develop community-responsive interdisciplinary computing solutions, addressing societal challenges and contributing to sustainable technological solutions that benefit communities globally.

IS Capstone Project

In CHED's CMO. No. 25 S. 2015, Capstone Project Bachelor of Science in Information System (BSIS) programs. This project function as terminal project requirements that would only demonstrate a student's comprehensive knowledge of the area of study and research methods used but also allow them to apply the concepts and methods to a specific problem in their area of specialization. *BS Information Systems students must complete a project such as business application development, or an Information Systems plans.*

An Information Systems Capstone Project focuses on business processes and the implications of introducing a Computing solution to a problem.

- Software Development
  - Software Customization
  - Information Systems development for actual client
  - Web Applications Development
  - Mobile Computing Systems
- IS Planning
  - Enterprise Resource Plan
  - Information System Strategic Plan
- Analysis and Design of a sufficiently complex business system

CCICT IS Capstone Project

In the CCICT- BSIS curriculum, Capstone Project is also known as a Capstone Project. This subject is the most critical evaluation of whether the previous knowledge and skills learned by the students from his/her freshman year to senior year are applied and integrated.

The students are encouraged to propose a **Technology Start-Up Idea** for a capstone project that is not existed or proposed by previous proponents/researchers. The proposed capstone project may fall under, but not limited to, any of the following categories:

- Cloud Computing (Saas, PaaS, IaaS)
- Social Media
- Internet of Things (IOT)
- Web Applications
- Mobile Apps (iOS, Android)

The proposed capstone project (Technology Start-Up Idea) may address the following areas, but not limited to:

- Agriculture
- Commerce
- Disaster management
- Education
- Environment
- Governance
- Health care
- Home
- Livelihood
- Media and Entertainment
- Lifestyle/ People-on-the-Go
- Power and Energy
- Social Sciences
- Telecommunications
- Tourism
- Transportation

Research Courses/Subjects

In the CCICT - BSIS, the Capstone Project is divided into two (2): Capstone Project 1 which is focused on the planning, analyzing, and designing of the proposed study/project; and Capstone Project 2 which is more focused on the development and implementation of the same study/project.

Capstone Project Teacher / Instructor

The Capstone Project Teacher is one handling/facilitating a class of 20 to 30 students in one or more Capstone Project 1 and 2 courses/subjects. He/she is responsible for the orientation of the students on what to expect of the course/class, conduct meetings, organize research-related events, create schedule Proposal Hearing and/or Oral Defense proceedings, and deliver grade. By default, he/she is considered a panel member in graded proceedings.

Duties and Responsibilities of the Capstone Project Teacher / Instructor

- Announce the Research or Capstone Project Areas or theme approved by the dean to the students;
- Conduct general / specific meetings with the students to discuss the Capstone Project Policies, Standards, Guidelines, and Deliverables and to allow the students to raise and clarify issues;
- Facilitate sound communication line(s) for and between the students, faculty members, advisers, panel members, and Dean.
- Select a Proposal Hearing and/or Oral Defense panel for each team;
- Schedule Capstone Project activities, such as the deadlines of deliverables and Proposal Hearing and Oral Defense sessions.
- Post schedules, Proposal Hearing and Oral Defense guidelines, requirements guidelines, and other announcements;
- Furnish every member of the Proposal Hearing and Oral Defense panel with all the necessary Capstone Project documents before the scheduled Proposal Hearing or Oral Defense respectively;
- File at least one copy of the Proposal Hearing and Oral Defense panel's evaluation (including revisions) and the Revised and Approved Deliverables at every stage of the Capstone Project.
- Submit collated grades of students under his/her section for that term.

Capstone Project Team Composition

Each Capstone Project Group must consist of three (3) or four (4) members who will take on at least one of the following roles:

Hustler - Project Manager

- *The Hustler optimizes marketing strategies in promoting the product or output.*
- The person in authority to manage a Capstone Project from start to finish. This includes leading the planning and the development of all Capstone Project deliverables.
- Responsible for the budget, work plan, and all Project Management Procedures (eg. Scope management, issues management, risk management, etc.)
- Responsible for the success of the entire activity.

Hacker - Software Engineer, Software Tester

- *The hackers will be the ones to build/develop the system or application.*
- The person who designs, writes, and test computer programs.

Hipster – Systems Analyst/ Database Designer/UI Designer

- The person who checks that all parts of the system are coordinated based on a complete and traceable Software Requirements and Specifications (SRS)
- The person who makes sure that the storyboard and the database design are complete and robust.
- *The hipster is in charge of making sure that there is a logical connection between the design and user requirements.*

Duties and Responsibilities of the Proponents/Researchers

- Keep informed of the Capstone Project and Research Guidelines and Policies.
- Keep informed of the schedule of Capstone Project and Research activities, required deliverables, and deadlines posted by the Instructor and Adviser.
- Submits on time all deliverables specified by the Instructor and Adviser.
- Submits on time all requirements identified by the Capstone Project and Research Oral Defense Panel during the Oral Defense.
- Submits on time the requirements identified by the Instructor and Adviser throughout the duration of the Capstone Project and Research.
- Schedules regular meetings with the Adviser throughout the duration of the Capstone Project and Research. The meetings serve as a venue for the proponent to report the progress of their work, as well as raise any issues or concerns.

Team Regrouping

Regrouping is allowed if less than three (3) members of the group remain. Should this happen, the group may be disbanded and members of this affected group may join other groups as long as the maximum number for each group is followed. However, if the remaining member(s) decide(s) to continue with his/her/their project, regrouping may not apply but with the consent of both the Project Leader, Capstone Project Adviser, Instructor, Program Chairperson and the College Dean. Revision of the scope may then be an option. The title/topic to be pursued will then be decided among the team members, Project Leader, Capstone Project Adviser, Instructor, Program Chairperson and the College Dean.

Project Advisers

Any full-time regular faculty member of the CCICT can be a project adviser as long as the project falls under his/her area of expertise or specialization. If full-time instructors are insufficient, part-time teachers can be assigned as a project adviser as long as he/she is teaching in the college.

Duties and Responsibilities

Project Adviser:

1. Ensures that the study proposed by the students conforms to the standard of the College and has immediate or potential impact on the research thrust of the university.

2. Guides the research/project students in the following tasks while in the proposal stage:
    a. Defining the research problems/objectives in clear and specific terms
    b. Building a working bibliography for the research
    c. Identifying variables and formulating hypothesis, if necessary
    d. Determining research design, population to be studied, research environment, instruments to be used and the data collection procedures

3. Meets the team regularly (at least once a month) to answer questions and help resolve impasses and conflicts.

4. Points out errors in the development work, in the analysis, or in the documentation. The adviser must remind the proponents to do their work properly.

5. Reviews thoroughly all deliverables at every stage of the project, to ensure that they meet the College's standards. The adviser may also require his/her project proponents to submit progress reports regularly.

6. Recommends the proponents for proposal hearing and oral defense. The project adviser ***should not*** sign the Hearing Notice (for Proposal Hearing and Oral Defense) if he/she believes that the proponents are not yet ready for proposal hearing and/or oral defense.

7. Clarifies points during the proposal hearing and oral defense.

8. Ensures that all required revisions are incorporated in the appropriate documents and/or software.

9. Keeps informed of the schedule of project activities, required deliverables and deadlines.

10. Recommends to the project defense panel the nomination of his/her project for an award.

11. As a project adviser, he/she is responsible to be:

    a. A provider
    b. An encourager
    c. A dictator
    d. A pushy boss e.
    A connector

Adviser Reassignment
In the event that the Adviser needs to leave the team, for valid reasons assessed by the Dean, such as long-term Leave of Absence and unavoidable circumstances among others, the group/team may be reassigned to another adviser. This may happen from Capstone Project 1 to Capstone Project 2.

Capstone Project Panel Composition

The capstone Project panel is composed of Chairman, Members, and Adviser.

- 1 Project Adviser
- 2 Panel Members (assigned by the Subject Teacher) 1 Content Expert (the Dean/Chair or his/her Representative)
- 1 Chairman (Dean or his/her Representative)

Panel Reassignment
In the event that the Chairman or any panel members needs to leave/excuse the proceedings, for valid reasons assessed by the Dean, such as Leave of Absence, Official Business, and unavoidable circumstances among others, the group/team may be reassigned to another equally competent panelist. This may happen from Capstone Project 1 to Capstone Project 2.

Capstone Project Timeline
The BSIS Capstone Project shall be completed within two (2) semester (Capstone Project 1 and Capstone Project 2)

Capstone Activities
The following are the important activities in the course of Capstone Project 1 and Capstone Project 2:

Capstone Project	Timeline
1. Enrollment	
2. Orientation	Prelim Period
3. Pitch Deck Preparation	Prelim Period
4. Lightning Pitch	Prelim Exam Week
5. Proposal Manuscript Writing	Midterm Period
6. System Planning	Midterm Period
7. System Analysis	Midterm Period

8. System Design — Semi-Final Period
9. Proposal Hearing — 3 weeks before the Final Exam
10. Startup Contests Application (if available)

Capstone Project 2 — **Timeline**

1. Enrollment
2. Orientation — Prelim Period
3. Capstone Project Advisers' Consultation — Prelim – Final Period
4. Manuscript Rewriting — Prelim – Final Period
5. System Planning and Analysis — Prelim Period
6. System Designing — Midterm Period
7. System Development — Midterm – Final Period
8. System Testing — Midterm – Final Period
9. Oral Defense — 3 weeks before the Final Exam
10. Startup Contests Application (if available)

Grading System

There will be different sets of grading sheets/rubrics to be employed for Capstone Project 1 and Capstone Project 2. It will consist of grading variables like deliverables, group grade, individual, adviser's grade, and even project management evaluation.

Capstone Project and Research 1

Title Defense		**85%**
	Adviser's Grade:	
	Attendance / Behavior / Professionalism	10
	Role Commitment/Contribution	
	Panels' Grade:	
	Team (Manuscript / System Design)	60
	Individual (Oral Presentation)	30
	Total	100
Deliverables		**15%**
	(Prelim, Midterm, Semi Finals)	100

Skills Test

This is an individual grade component that assesses a student's problem-solving, critical thinking, and technical/programming skills to ensure that he or she is prepared to complete his or her research project and subsequently enter to the industry. The problem entails both backend and frontend aspects and modification, as well as extensive understanding of file/database management. The exam does not necessitate the use of any specific frontend or backend tools or programs.

Adviser's Grade

The adviser will grade each member of the team based on behavior, professionalism, significance, contribution, and commitment/competence to the role assigned.

Panel's Grade

The grade of the deliverables covers the requirements to be submitted during prelim, midterm and semi-finals period. Each period, the group must submit and comply the requirements before the deadline that is set by the professor/instructor of the subject.

Deliverables

The grade of the deliverables covers the requirements to be submitted during prelim, midterm and semi-final period. Each period, the group must submit and comply the requirements before the deadline that is set by the professor/instructor of the subject.

Group Grade

Manuscript	**60 points**
Preliminary Pages Table of contents is consistent Acknowledgment is brief and formal	**5**
Chapter 1 The Problem and Its Scope Introduction Rationale of the Project is intact and provides a clear overview of the entire Research/Capstone Project The Problem Statement of the Problem/Objectives of the Project is 'SMART' - Specific, Measurable, Achievable, Realistic, and Timely Scope and Limitations are clearly defined Significance of the study section enumerates and elaborates operationally all the beneficiaries of the system/capstone	**15**
Chapter 2	
Review of the Related Literature and Studies are recent and relevant. Anchor provides solid background of the Research/Capstone Project. Auxiliary theories/algorithm are evident. Sources are appropriately cited and noted. Related studies are relevant and includes global and local scope.	**15**

Chapter 3	
Methodology strictly follows the SDLC Methodology includes project management techniques appropriate for the chosen Research / Capstone Project Requirements Specification is more or less complete and answers the objectives Design tools used are relevant and appropriate which should be based on requirements Development plan is concrete and should be consistent with the Design Testing Techniques to be used should assess all aspects of the developed Research / Capstone Project Implementation Plan should be aligned with the objectives	**15**
Final Pages	
Findings and Conclusions are attuned with the objectives Recommendations are feasible and practical References should be APA Format Appendices are relevant and help support the principal content	**5**
Manuscript Mechanics	
Organization and Fluidity of ideas are apparent Formatting and layout are consistent All parts of the manuscript should be grammatically correct	**5**

Individual Grade

Oral Examination	**30 points**
Comprehensiveness of the Answer/Ideas	20
Contribution/Support to the Team	5
Delivery/Command of the English Language	5

Verdicts

There will be three (3) possible verdicts after the Proposal Hearing. The verdict is a unanimous decision among the three members of the Capstone Project Oral Defense panel. Once issued, it is final and irrevocable.

CAPSTONE PROJECT 1 | VERDICT	
	APPROVED WITH MINOR REVISIONS. Minor revisions are necessary but they do not have to be presented in front of, and checked by all panelists.
	APPROVED WITH MAJOR REVISIONS. Major revisions are necessary but they do not have to be presented in front of, and checked by all panelists. (50-9) -> Major (70-100)→Minor
	DISAPPROVED. The proponents failed to propose a researchable or scholarly project. This verdict will not compel the panelists to give numeric grades. Team members who will get this verdict during the Proposal Hearing will be given a **5.0 FINAL GRADE.**

Capstone Project 2

Skills Test (Individual)	20
Role Commitment/Project Management	
◦ Project Managers grades his/her team members	10
◦ Adviser grades the Project Manager	
Panels' Grade:	
Output (Developed System) Group Grade	
◦ Title = Objectives = Output	20
◦ Scope = Modules = Output	20
◦ System = Methods = Output	10
Individual (Oral Presentation)	20
Total	**100**

Panels Grade

The panel's quantitative rating is composed of two (2): **group grades** and **individual grades**. Group grade covers the system output that is coherent with the title of the study, the set of objectives, the identified modules which includes methods used to deliver the goals and deliverables of the system. Individual grade, covers the oral presentation and the relevance of his/her answer/s to question/s. Each member of the panel quantitative ratings comes with a directly-proportional qualitative verdict. Both quantitative ratings and qualitative verdicts will be
collated by the chairman of the panel.

Group Grade

Capstone Project | Output | 50 points	
The output should be consistent with the title and objectives as defined during the proposal stage	20
The output should be consistent with the scope and modules as defined during the proposal stage	20
The output should be consistent with the kind of system and the methods employed as defined during the proposal stage.	10

Individual Grades

Capstone Project and Research 2 | Output | 20 points	
◦ Comprehensiveness of the Answer/Ideas	10
◦ Contribution/Support to the Team	5
◦ Delivery/Command of the English Language	5

Verdicts

CAPSTONE PROJECT 2 | VERDICT	
	ACCEPTED WITH REVISIONS. Revisions are necessary but they do not have to be presented in front and check by all panelists.
	RE-ORAL DEFENSE. Another Oral Defense session, in which all panelists must be present, is necessary to further clarify the objectives and scope of the capstone project.
	NOT ACCEPTED. The proponents failed to achieve the objectives of the research established in the proposal. This verdict will not compel the panelists to give numeric grades. Team members who will get this verdict will be given a **5.0 FINAL GRADE.**

Capstone Project 1 Guidelines:

1. **Enrolment.**
   Prior to the enrolment of the subject, pre-requisites should have complied. Only officially-enrolled Capstone Project 1 and Capstone Project 2 students are allowed to join a group and proceed with pertinent activities thereafter.

2. **Major Subjects.**
   Subject for which the student has chosen the Core Courses (Common Courses, Professional Courses, Professional Electives & Domain Courses) in the respective program.

3. **Orientation.**
   On the first week of classes, a general orientation will be conducted where this Capstone Project Manual will be discussed. All students will then e required to secure the Oath of Confirmation pertaining to the receipt of copy and discourse about the said manual (Refer to *Appendix A: Capstone 1 Oath of Confirmation Form*). This form will be submitted on the set deadline/due date.

4. **Grouping.**
   Part of the general orientation will be the grouping activity where all students across and regardless of section/offer code are given the privilege to form a team of 3 to 4 members. Upon the team organization, each group will have to secure the Team Composition Form containing the important profile of each member and the corresponding role assignments (Refer to *Appendix B: Capstone Project Team Composition Form*). This form will be submitted on the set deadline/due date.

5. **Pitch Deck Preparation.**
   Immediately after the team is organized, they will be required to prepare a pitch deck in the form of a Powerpoint presentation (Refer to *Appendix W: Pitch Deck Outline*) ready for the scheduled lightning pitch.

6. **Initial Review of Related Literature/Studies**.
   Along with the Pitch Deck, the team will have to start reviewing relevant literatures and studies for each topic.

7. **Lighting Pitching.**
   The Subject Teacher will then schedule the lightning pitch for each team. It is aimed at selecting a final topic of interest for the team. Once a topic of interest is selected and agreed upon by all, the team will then secure the working title form.

8. **Advisers' Assignment.**
   The Dean and the Faculty will then meet and decide/appoint/assign each team's adviser based on expertise and other objective considerations. The CCICT may also give the team the discretion to choose their desired adviser subject to Dean's approval. The Working Title Forms will then be signed by the identified adviser of each team.

9. **Adviser-Team Consultation.**
   The team will then start doing formal consultations with their adviser. Each team is required at least five (5) consultation cycles imposing the extent of completion of deliverables. (Refer to *Appendix G. Capstone 1 Consultation Logs Form*). The adviser and the project manager will affix their signatures in each formal consultation. The said form will be required in the processing of the Proposal Hearing Notice. Hence, the adviser will be made to endorse the team whether or not they are allowed to propose.

10. **Proposal Presentation Preparation.**
    The proposal hearing presentation must be prepared well by the team. It must strictly follow the set outline and rules. (Refer to *Appendix J. Presentation Outline and Guidelines*).

11. **Proposal Hearing.**
    Proposal Hearing and Mechanics will have to be followed and strictly adhered. (Refer to *Appendix K. Proposal Hearing Rules and Mechanics*).

12. **Proposal Manuscript Revisions.**
    The team and adviser will ensure that all the recommendations will have been incorporated in the Revised Proposal Manuscript with the corresponding title page.

13. **Proposal Approval Sheet Routing.**
    The team then routes the Proposal Hearing Approval Sheet to all required Signatories.

14. **Submission of Proposal Softbound and CD/DVD.**
    After all the signatories will have signed the approval sheet, the team then has to have the entire proposal manuscript soft- bounded. The color of the softbound is **RED**. The softbound together with the Proposal CD/DVD will be submitted to the College. (Refer to *Appendices U and V. Capstone Project CD/DVD Labeling format and Appendix Capstone CD/DVD Contents Guide*).

Capstone Project 2 Guidelines:

1. **Orientation.**
On the first week of classes, a general orientation will be conducted where this Capstone Project Manual will be discussed again. All students will then be required to secure the Oath of Confirmation pertaining to the receipt of copy and discourse about the said manual (Refer to *Appendix O: Capstone 2 Oath of Confirmation Form*). This form will be submitted on the set deadline/due date.

2. **Capstone Project Advisers' Forum.**
Each team adviser must meet his/her teams in one' setting. General and specific issues and concerns must be reported and settled. Furthermore, if there are changes or the Dean and Subject Teacher approve team regrouping, a new duly signed copy of the Team Composition Form must be secured and submitted to the Subject Teacher.

3. **Finalization of the List of Modules.**
The Project Manager and his/her team must finalize the List of Modules. The adviser must check that the same list is anchored from the title, objectives and scope.

4. **Adviser-Team Consultation.**
The team will continue doing formal consultations with their adviser. The team is required of at least three (3) prototype consultation cycles imposing the extent of completion of modules (Refer to *Appendix P. Capstone 2 Consultation logs*). The adviser and the project manager will affix their signatures in each formal consultation. **The adviser will be made to endorse whether or not they are allowed to defend their project or not.**

5. **Panel-Team Consultation.**
The team can, from time to time, consult any member of the Panel to whom they proposed/defended for clarifications of their recommendations. The panel member must ensure that academic integrity is observed and therefore must not be biased in any form/aspect.

6. **Oral Defense Manuscript Finalization.**
The manuscript for Capstone 2 must be completed and finalized by the team and their adviser. (Refer to *Appendix Q. Capstone 2Manuscript Outline).*

7. **Oral Defense Presentation Preparation.**
The oral defense presentation must be prepared well by the team. It must strictly follow the set outline and rules.

8. **Oral Defense**. During the proceedings, the adviser and the team will take note of all the recommendations of the Oral Defense panel.

9. **Oral Defense Manuscript Revisions.**
The team and adviser will ensure that all the recommendations will have been incorporated in the Revised Oral Defense Manuscript with the corresponding title page. The same title page template will be used for the hardbound cover page.

10. **System Output Revisions.**
The team then implements all the recommendations not only the manuscript but also the important system or output. The said revisions will be bases of' the panel member(s) in the signing of the Approval Sheet.

11. **Oral Defense Approval Sheet Routing.**
The team will route the Oral Defense Approval Sheet together with the revised system output to all required signatories. (Refer to *Appendix S Capstone 2 Approval Sheet Template*).

12. **Submission of Oral Defense Hardbound and CD/DVD.**
After all the panel members and signatories have signed the approval sheet, the team then has to have the entire oral defense manuscript hard bounded. The color of the hardbound is **RED** with **GOLD TEXT.** The hardbound together with the CD/DVD will be submitted to the Subject Teacher.

13. **Research/Capstone Project Colloquium.**
The college will then select Capstone Projects/Researches worthy to be highlighted in either a college-organized colloquium and/or externally-organized research events.

Appendices

Appendix A
Capstone Project 1
Oath of Confirmation

Date: _____

OATH OF CONFIRMATION

This is to confirm that I, _____, a BSIS-3 student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

(Signature of Student over Printed name)/ Date

Date: _____

OATH OF CONFIRMATION

This is to confirm that I, _____, a BSIS-3 student, currently enrolled in the course of Capstone Project 1, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

_____

(Signature of Student over Printed name)/ Date

Appendix B
Capstone Project Team Composition Form

Name	Signature	Role Assignment	Email	Contact #
1.				
2.				
3.				
4.				

Appendix C
Acceptance of Capstone Adviser

College of Computer, Information and Communications Technology

Acceptance of Capstone Adviser

This is to certify that I am accepting the ___ *(Name of the Team)* ___ to be their Capstone Project adviser in the degree of ___ *(Degree Program)* ___.

_____

(Signature of Adviser over printed name)

Date: _____

Noted by:

_____

(Signature of Subject Teacher over printed name)

Date: _____

Appendix D
Capstone Project Business Model Canvas Template



Appendix E
Capstone Project Comparative Matrix Template

Features	Application 1	Application 2	Application N	Your Application

Appendix F
Research Working Title Form

Name of the Proponents	
1.	
2.	
3.	
4.	
Proposed Research Title:	
Date:	Date:
Submitted by:	Noted and Approved by:
(Signature of Project Leader over Printed Name)	(Signature of Project Leader over Printed Name)

Appendix G.
Capstone Project 1 Consultation Logs Form

Research/Capstone Project Title

Name of the Proponents

Name of the Group

1st Consultation : CHAPTER 1 must be completely delivered for adviser's evaluation

Date	PM's Signature	Adviser's Signature

2nd Consultation : CHAPTER 1 and 2 must be completely delivered for adviser's evaluation

Date	PM's Signature	Adviser's Signature

3rd Consultation : **CHAPTER 1, 2 and 3 must be completely delivered for adviser's evaluation**

Date	**PM's Signature**	**Adviser's Signature**

4th Consultation : **CHAPTER 1, 2, 3 and including initial and finals pages must be completely delivered for advisers' evaluation**

Date	**PM's Signature**	**Adviser's Signature**

This is to certify that I have been regularly consulted by my advisees; have reviewed their manuscript of the above-stated study. As their adviser, I therefore submit them ready for Proposal Hearing as the Chapters 1, 2, and 3 and the pertinent parts of their manuscript are complete

(Signature of Adviser over printed name)

Appendix H
Capstone Project 1 Manuscript Outline

- TITLE PAGE
- ACKNOWLEDGMENT
- DEDICATION
- TABLE OF CONTENTS
- LIST OF TABLES
- LIST OF FIGURES
- CHAPTER I - INTRODUCTION
  - Rationale of the Study
  - Objectives of the Study
  - Significance of the Study
  - Scope and Limitation of the Study
  - Flow of the Study
  - Definition of Terms
- CHAPTER II - REVIEW OF RELATED LITERATURE AND STUDIES
  - Related Literature
  - Related Studies
- CHAPTER III - RESEARCH METHODOLOGY
  - Software Engineering Methodology
  - Planning Phase
    - Business Model Canvas
    - Program Workflow
    - Gantt Chart
    - Functional Decomposition Diagram
  - Analysis Phase
    - Use-Case Diagram
    - Storyboard
    - Database Design
      - Entity-Relationship Diagram
      - Data Dictionary
    - Network Design
      - Network Model
      - Network Topology
  - Development Phase
    - Technology Stack Diagram
      - Software Specification
      - Hardware Specification
    - Program Specification
      - List of Modules
- REFERENCES
- CURRICULUM VITAE
- APPENDICES

Appendix I
Capstone Project Manuscript Mechanics

The Capstone Project Manuscript must follow the following format and technicalities:

a. Paper
   i. Size : 8.5 x 11
   ii. Orientation : Portrait (except for special diagrams)
   iii. Substance : 20
b. Spacing
   i. 1.5 inches
c. Indention
   i. 1 tab
d. Margins
   i. Top : 1 inch
   ii. Left : 1.5 inch
   iii. Bottom : 1 inch
   iv. Right : 1 inch
   v. Header : 0.5 inch
   vi. Footer : 0.5 inch
e. Font
   i. Sizes
      a. Heading 1 : 12
      b. Heading 2 : 12
      c. Heading 3 and Content : 11
   ii. Type: Strictly Times New Roman
   iii. Color: Black
f. Pagination
   i. Top Right
   ii. No page shown on first page of every chapter
g. Page Breaks
h. Page break is only used when starting a new chapter
i. Sample Layout for Tables

Table Number
Table Title

j. Sample Layout for Figures



Figure Number: **Figure Title (bold, underlined)**

Appendix J
Capstone Project 1 Presentation Outline and Guidelines

Presentation Slides Outline (Proposal Hearing)

- Introduction
- Concept/Explainer Video
- Title Page (Includes the Title, Proponents with Role (PM will introduce to the Panel)

- Chapter 1
  - Rationale (Verbatim)
  - Objectives of the Study/Project (Verbatim)
  - Significance of the Study/Project (just the beneficiaries, elaborate orally)

- Chapter 2

  - Related Literature (bulleted terms only, explain and elaborate orally)
  - Comparative Matrix

- Chapter 3

  - Environment (Optional)
  - Software Engineering Methodology

  - Planning Phase
    - Business Model Canvas
    - Program Workflow
    - Gantt Chart
    - Functional Decomposition Diagram

  - Analysis Phase
    - Use-Case Diagram
    - Storyboard
    - Database Design
    - Entity-Relationship Diagram
    - Data Dictionary
    - Network Design

  - Development Phase
    - Technology Stack Diagram
    - Software Specification
    - Hardware Specification
    - Program Specification
      - List of Modules

  - Testing Plan (bulleted terms only, explain and elaborate orally)

Appendix K
Proposal Hearing Rules and Mechanics

Should be strictly followed both panel and presentations

I.	Preparations (Initial Preparations)	5 min
II.	Proposal Hearing Proper (Presentation)	25 min
III.	Question and Answer (Q & A) • Involves the panel's clarification and cross- examination	40 min
IV.	Panel Deliberation • Grades consolidation • Verdict deliberation	10 min
V.	Verdict Presentation • The chairman announces the group average grade, but not the individual grade	5 min
VI.	FINALIZATION / FINAL ADJUSTMENT	5 min
	TOTAL	90 mins

•

Appendix L
Capstone 1 Project Manuscript Title Page Template

TITLE TITLE TITLE TITLE TITLE TITLE TITLE TITLE TITLE TITLE
TITLE TITLE TITLE TITLE TITLE TITLE TITLE TITLE
TITLE TITLE TITLE TITLE TITLE TITLE

(Must be inverted pyramid form, all caps)

A Capstone Project Presented to the Faculty of
College of Computer, Information and Communications Technology Department
Cebu Technological University - Main Campus

In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information System

By

Juan Dela Cruz
Pedro Reyes Jo
Rizal
Apo Mabini

Bill Gates
Adviser

May 2025

Appendix M
Capstone Project 1 Approval Sheet Template

APPROVAL SHEET

The Capstone Project entitled **Municipality of Argao Public Vehicle Information System** prepared and submitted by Juan Dela Cruz and Peter Reyes has been examined and is recommended for approval and acceptance.

Recommended:

BILL GATES, Ph.D

Adviser

APPROVED by the Committee on PROPOSAL HEARING with a verdict of _____ on _____.

CHAIRMAN C. CHAIRMAN

Chairman

PANEL A. MEMBER

Member

PANEL B. MEMBER

Member

ACCEPTED and APPROVED in partial fulfillment of the requirements in Bachelor of Science in Information System

ASSOCIATE A. DEAN
Associate Dean, CCICT

DEAN O. DECOLLEGE
Dean, CCICT

Appendix N
List of Modules

Programmer	Modules	User 1	User 2	User 3
Programmer	**Module Name 1**			
	1. Unit 1			
	2. Unit 2			
	3. Unit 3			
	4. Unit N			
No. of Points				
Programmer 2	**Module Name 2**			
	1. Unit 1			
	2. Unit 2			
	3. Unit 3			
	4. Unit N			
No. of Points				
Total Number of Modules				

Example:

Programmer	Modules	Owner	Manager	Driver
Programmer 1	**Client Management**			
	1. Create	*	*	
	2. Retrieve	*	*	*
	3. Update	*	*	
	4. Delete	*	*	
No. of Points		1		
Programmer 2	**Delivery**			
	1. Create	*	*	
	2. Retrieve	*	*	*
	3. Update	*	*	
	4. Delete	*	*	
	5. Assign Driver	*	*	
No. of Points		1		
Total Number of Modules		2		

Appendix O
Capstone 2 Oath of Confirmation

Date: _____

OATH OF CONFIRMATION

This is to confirm that I, _____, a BSIS-3 student, currently enrolled in the course of Capstone Project 2, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

_____

(Signature of Student over Printed name)/ Date

Date: _____

OATH OF CONFIRMATION

This is to confirm that I, _____, a BSIS-3 student, currently enrolled in the course of Capstone Project 2, have aptly received a copy; religiously read and understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the Capstone Project Manual document implemented by the Subject Teacher.

This is also to affirm that said guidelines were judiciously discussed and explicitly elaborated in a forum conducted by the Subject Teacher.

CONFORME:

_____

(Signature of Student over Printed name)/ Date

Appendix P
Capstone Project 2 Consultation Logs Form

Research/Capstone Project Title

Name of the Proponents

Total Number of Modules: **Noted by Subject Teacher:**

Prototype	Date of Consultation	# of Modules Fully Implemented	#of Modules Partially Implemented	Running Score	%	PM Signature	Adviser Signature
1st Prototype 30% - 50% of the modules must be running	Remarks:						
2nd Prototype 51% - 70% of the modules must be running	Remarks:						

	Remarks:						
3rd Prototype 71% - 100% of the modules must be running							

This is to certify that I have been regularly consulted by my advisees; have reviewed their system output as well as the required manuscript of the above-stated study. As their adviser, I, therefore, submit them ready for **Oral Defense** as their third prototype is within the required percentage.

_____

(Signature of Adviser over printed name)

Appendix Q
Capstone 2 Manuscript Outline

- Title Page
- ACKNOWLEDGMENT
- DEDICATION
- TABLE OF CONTENTS
- LIST OF TABLES
- LIST OF FIGURES
- CHAPTER I - INTRODUCTION
  - Rationale of the Study
  - Objectives of the Study
  - Significance of the Study
  - Flow of the Study
  - Definition of Terms
- CHAPTER II - REVIEW OF RELATED LITERATURE AND STUDIES
  - Related Literature
  - Related Studies
- CHAPTER III - RESEARCH METHODOLOGY
  - Software Engineering Methodology
  - Planning Phase
    - Business Model Canvas
    - Program Workflow Gantt
    - Chart
    - Functional Decomposition Diagram
  - Analysis Phase
    - Use-Case Diagram
    - Storyboard Database
    - Design
      - Entity-Relationship Diagram
      - Data Dictionary
    - Network Design
      - Network Model
      - Network Topology
  - Development Phase
    - Technology Stack Diagram
      - Software Specification
      - Hardware Specification
    - Program Specification
      - List of Modules
    - Testing Phase
      - Unit Testing
      - Integration Testing
      - Alpha Testing
      - Acceptance Testing
  - Implementation/Deployment Phase
    - Software Specification Hardware
    - Specification
    - User-Guide
    - Installation Guide
    - Roadmap
- CONCLUSION
- RECOMMENDATIONS
- REFERENCES
- CURRICULUM VITAE
- APPENDICES

Appendix R
Capstone Project 2 Oral Defense Presentation Outline and Guidelines

Presentation Slides Outline (Oral Defense)

- Concept/Explainer Video
- Title Page (includes title, Proponents with Role)
- Chapter 1
    - Objectives of the Study/Project (bulleted, verbatim)
    - Scope and Limitations (bulleted terms, explain/elaborate orally)
- Chapter 3
    - Functional Decomposition Diagram
    - Use Case Diagram
    - Entity Relationship Diagram
    - Technology Stack
    - Network Design
    - Project Roadmap
- Conclusions
- Recommendations
- System Demo

Appendix S
Oral Defense Rules and Mechanics

I.	Preparations (Initial Preparations)	5 min
II.	Defense Proper (Presentation)	25 min
III.	System Demo (Q & A) • Involves the panel's clarification and cross-examination	40 min
IV.	Panel Deliberation • Grades consolidation • Verdict deliberation	10 min
V.	Verdict Presentation • The chairman announces the group average grade, but not the individual grades.	5 min
VI.	FINALIZATION / FINAL ADJUSTMENT	5 min
	TOTAL	90 mins

Appendix T
Capstone Project 2 Manuscript Title Page Template

CTU

LIGTASUNA: WEB AND MOBILE – BASED FIRST AID APPLICATION FOR VICENTE SOTTO MEMORIAL MEDICAL CENTER

FEBRUARY 2022

Appendix S. Capstone 2 Manuscript Title Page Template

CEBU TECHNOLOGICAL UNIVERSITY – MAIN CAMPUS
M.J. Cuenco Ave, Cor R. Palma St., Cebu City

LIGTASUNA: WEB AND MOBILE – BASED FIRST AID APPLICATION FOR VICENTE SOTTO MEMORIAL MEDICAL CENTER

GIBE S. TIROL
BELL S. CAMPANILLA
PET ANDREW P. NACUA
ANGELBERT P. MAGHANOY

Submitted in Partial Fulfillment
of the Requirements for the Degree

Bachelor of Science in Information System

February 2022

Appendix U
Capstone 2 Approval Sheet Template

APPROVAL SHEET

The Capstone Project entitled **Municipality of Argao Public Vehicle Information System** prepare and submitted by Juan Dela Cruz and Peter Reyes has been examined and is recommended for approval and acceptance.

Recommended:

BILL GATES, Ph.D
Adviser

APPROVED by the Committee on ORAL DEFENSE with a verdict of _____ on _____.

CHAIRMAN C. CHAIRMAN
Chairman

PANEL A. MEMBER
Member

PANEL B. MEMBER
Member

ACCEPTED and APPROVED in partial fulfillment of the requirements in Bachelor of Science in Information Systems

ASSOCIATE A. DEAN
Associate Dean, CCICT

DEAN O. DECOLLEGE
Dean, CCICT

Appendix V
Capstone Project CD/DVD Labelling Format

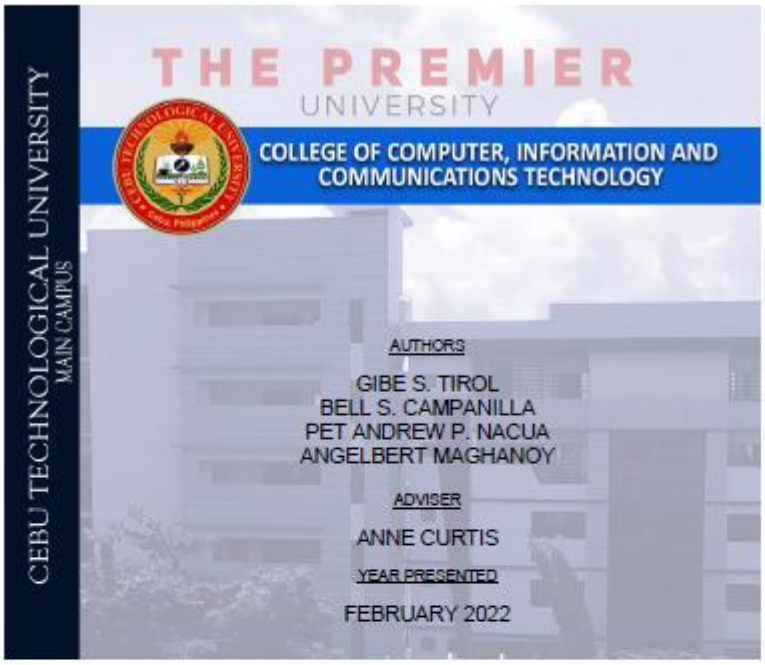


Size:

H: 4.8"

W: 5.5"

Appendix W
Capstone Project CD/DVD Case Cover Format



Steps:

1. Insert picture
2. Select the picture and go to format menu
3. Click the drop down arrow located at the bottom right part of Size Option Menu
4. Uncheck the following:
    * Lock aspect ratio
    * Relative to original picture size
5. Set the picture size to :
    H:4.8"
    W: 5.5"
6. Edit the text

Appendix X
Pitch Deck Outline

Slide 1	Title	Provide company name, and startup idea title, and contact details such as an address, email, and cell number.
Slide 2	Problem/Opportunity	Describe the pain point that you're alleviating, or the pleasure you're providing
Slide 3	Value Proposition	Explain the value of the pain you alleviate or the value of the pleasure you provide.
Slide 4 **I**	Underlying Magic	Describe the technology, secret sauce, or magic behind your product. The less text and the more diagrams, schematics, and flowcharts the better. If you have a prototype or demo, this is the time to transition to it.
Slide 5	Business Model	Explain who has the money temporarily in his pocket and how you're going to get it into yours. Simply put, describe who your customers are, and your revenue streams.
Slide 6	Go-To-Market Plan	Explain how you will acquire and retain your customers
Slide 7	Competition Analysis	Provide a complete view of the competitive landscape. Describe who is your toughest competitor in terms of strengths and weaknesses, and what makes your start-up idea/product leader.
Slide 8	Management Team	

Appendix Y. Capstone Project & Research (Change Advisory Form)
Appendix Z. Capstone Project & Research (Proposal Sheet)

SYSTEM PROPOSAL SHEET

Title of the Project:	Name of Proponents:
Sales and Inventory System of CCICT Company	James Reid Nadine Lustre Enrique Gil Liza Soberano
Semester:	Academic Year:

Brief Description of the System:

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu.
Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Aenean commodo ligula eget dolor. Aenean massa. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Donec quam felis, ultricies nec, pellentesque eu, pretium quis, sem. Nulla consequat massa quis enim. Donec pede justo, fringilla vel, aliquet nec, vulputate eget, arcu.

Statement of the Problem:

This research study designs and develops an automated Sales and Inventory System for LJDS Company.

Specifically, it answers the following problems:

1. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
2. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
3. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
4. Lorem ipsum dolor sit amet, consectetuer adipiscing elit; and
5. Lorem ipsum dolor sit amet, consectetuer adipiscing elit.

Objectives of the System:

The research study designs and develops an automated Sales and Inventory System for LJDS Company.

Specifically, it aims to:

1. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
2. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
3. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
4. Lorem ipsum dolor sit amet, consectetuer adipiscing elit; and
5. Lorem ipsum dolor sit amet, consectetuer adipiscing elit.

Scope and Limitation of the System:

The scope focuses on the Sales and Inventory System of LJDS Company.Specifically the system is capable of:

1. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   1.1 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   1.2 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   1.3 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   1.4 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
2. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   2.1 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   2.2 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   2.3 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   2.5 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
3. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   3.1 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   3.2 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
4. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   4.1 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
   4.2 Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
5. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;

However, the system is not capable of he following:

1. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
2. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
3. Lorem ipsum dolor sit amet, consectetuer adipiscing elit;
4. Lorem ipsum dolor sit amet, consectetuer adipiscing elit; and

Noted By:

Instructor

Approved By:

Angelbert P. Maghanoy
Program Chair - BSIS

References

CHED, (2013). *National Higher Education Research Agenda 2*. Retrieved from www.ched.gov.ph

CHED, (2015). *CHED Memorandum Order (CMO) No. 25. Revised Policies, Standards, and Guidelines for Bachelor of Science in Computer Science (BSCS), Bachelor of Science in Information Systems (BSIS), and Bachelor of Science in Information Technology (BSIT) programs*. Retrieved from www.ched.gov.ph

Niñal, M.M. and Satorre, S.B, (2015), *College of Computer Studies: Students' Capstone Project Manual*.